MPDD-AVG: Multimodal Personality-Aware Depression Detection via Audio-Visual Interview and Gait Analysis

Changzeng Fu, Yiming Zhang, Ming Shang, Jiadong Wang, Shiqi Zhao, Wenbo Shi, Alexis Meneses, Zhaojie Luo, Chaoran Liu, Yuichiro Yoshikawa, Hiroshi Ishiguro, and Björn Schuller

***Abstract*—Despite growing interest in automated depression detection, existing methods predominantly rely on conversational modalities while neglecting psychomotor behavioral signatures such as gait, and rarely incorporate individual-level factors including personality traits and demographic characteristics into their modeling frameworks. To bridge these gaps, the MPDD-AVG 2026 challenge introduces a multimodal benchmark integrating audio-visual interview data, wearable gait sequences, and rich personality and demographic annotations across two age cohorts. In this paper, we present the official baseline system employing a multimodal fusion architecture that encodes audio-visual cues, spectral-temporal gait features, and personality-conditioned textual embeddings derived from Big Five traits and demographic attributes. We evaluate the baseline on the MPDD-AVG benchmark across the Elderly and Young cohorts under three sub-tracks: Audio-Visual with Personality (A-V+P), Audio-Visual-Gait with Personality (A-V-G+P), and Gait with Personality (G+P), under both PHQ-9 regression and binary/ternary classification settings. Results demonstrate that gait incorporation yields notable gains, particularly in binary classification, while the proposed personality-aware design provides a personalized modeling framework for depression assessment, highlighting the importance of psychomotor and individual-difference modeling for robust multimodal depression detection.**

***Keywords*—**Personality-Aware, Multimodal Depression Detection, Gait Analysis

I. Introduction

DEPRESSION is a leading cause of global disability, imposing substantial societal and economic burdens on millions worldwide [1]. The advent of specialized benchmarks such as AVEC [2] and DAIC-WoZ [3] has significantly advanced audio-visual (A-V) depression detection, yet two critical limitations persist in existing methodologies.

First, current paradigms largely overlook *psychomotor retardation*, a hallmark objective sign of clinical depression [4] that manifests in ambulatory behaviors such as gait kinematics [5]. By confining observations to static interview settings, existing methods fail to capture this clinically established behavioral dimension, leaving a structurally important signal unexploited in computational frameworks. Second, prevailing models assume a uniform behavior-to-symptom mapping, fundamentally neglecting inter-individual heterogeneity driven by personality traits [6], demographic backgrounds, and physical comorbidities [7]. This limitation is particularly pronounced in late-life depression (LLD), where depressive manifestations are closely intertwined with underlying medical burdens and age-specific contextual factors [8].

To bridge these gaps, we introduce the **MPDD-AVG Challenge 2026**, which advances a holistic Audio-Visual-Gait paradigm for personality-aware depression detection. MPDD-AVG systematically integrates semi-structured interview data with spontaneous ambulatory phenotypes captured via wearable inertial measurement unit (IMU) sensors, and is organized into two age-specific cohorts: MPDD-Young and MPDD-Elderly (110 subjects each), enabling explicit modeling of age-sensitive and context-dependent symptom manifestations. Critically, MPDD-AVG provides granular individual-difference annotations to support personalized computational modeling, including PHQ-9 severity scores, Big Five-10 personality traits, native region demographics for the youth cohort, and socioeconomic and somatic disease annotations for the elderly cohort. As summarized in Table I, this rich annotation scheme empowers participants to investigate the complementary diagnostic value of continuous gait kinematics over static A-V cues, modulated by physiological and contextual priors. The challenge comprises two age-specific tracks, each featuring three sub-tracks: A-V+P, A-V-G+P, and G+P.

In this paper, we present the official baseline system for the MPDD-AVG 2026 challenge, providing a foundational framework and rigorous benchmark for participants. Our contributions are summarized as follows:

- We introduce the MPDD-AVG dataset, the first benchmark to jointly incorporate audio-visual, gait, and personality annotations for depression detection across two demographically distinct age cohorts.
- We propose a multimodal fusion baseline that integrates audio-visual cues, spectral-temporal gait features, and personality-conditioned textual embeddings, establishing competitive reference performance across all challenge tracks.
- We empirically demonstrate that gait modality incorporation yields consistent performance gains across multiple settings, while the proposed framework explicitly supports personality-aware depression modeling through

Changzeng Fu is with Northeastern University, China, and Osaka University, Japan. E-mail: fuchangzeng@qhd.neu.edu.cn
Yiming Zhang, Ming Shang, Shiqi Zhao, and Wenbo Shi are with Northeastern University, China.
Alexis Meneses is with Pontificia Universidad Católica del Perú, Peru.
Zhaojie Luo is with Southeastern University, China.
Chaoran Liu is with National Information Institute, Japan.
Yuichiro Yoshikawa and Hiroshi Ishiguro are with Osaka University, Japan.
Jiadong Wang and Björn Schuller are with the Technical University of Munich, Germany.

TABLE I
COMPARISON OF REPRESENTATIVE DATASETS FOR DEPRESSION-RELATED MULTIMODAL ANALYSIS.

Dataset	A	V	Gait (IMU)	Depression	Personality	Lifespan Cohorts	Comorbidity
DAIC-WoZ [3]	✓	✓	-	✓	-	-	-
Pittsburgh [9]	✓	✓	-	✓	-	-	-
D-Vlog [10]	✓	✓	-	✓	-	-	-
MODMA [11]	✓	-	-	✓	-	-	-
AMIGOS [12]	-	✓	-	-	✓	-	-
MPDD 2025 [13]	✓	✓	-	✓	✓	✓	✓*
MPDD-AVG (Ours)	✓	✓	✓	✓	✓	✓	✓*

Note: * Comorbidity annotations are specifically acquired for the elderly cohort.

individualized contextual representations, highlighting the importance of psychomotor and individual-difference modeling for robust depression detection.

II. DATASET

Despite the burgeoning interest in multimodal approaches for depression detection, existing methodologies exhibit notable limitations. While pioneering benchmarks like Pittsburgh [9], D-Vlog [10], CMDC [14], and MODMA [11] have advanced the field, most existing approaches predominantly rely on conversational or interview-based modalities (audio-visual), while ambulatory behavioral signatures such as gait characteristics, though recognized as important clinical indicators of psychomotor symptoms, remain largely underexplored. Moreover, many methods establish direct correlations between behavioral signals and depression levels without explicitly modeling individual differences, failing to account for the nuanced variations influenced by personality traits and demographic factors. While affective computing datasets such as AMIGOS [12] incorporate personality-related signals, they are not designed to jointly model depression, gait, and cohort-specific clinical context in a unified benchmark.

To bridge these gaps, this paper presents a comprehensive baseline model for the MPDD-AVG Challenge, which introduces a novel benchmark integrating semi-structured interviews with continuous gait monitoring. Our work directly addresses the aforementioned limitations. First, we propose a unified multimodal baseline framework that uniquely combines audio, visual, and gait data, enabling a holistic analysis spanning cognitive-linguistic, affective-paralinguistic, and psychomotor domains. Second, to effectively model individual heterogeneity, our framework incorporates personality-derived textual embeddings and demographic information.

Compared to prior works, our baseline is tailored to the unique characteristics of the MPDD-AVG dataset, effectively handling the heterogeneity of multi-source data (interview-based A/V and sensor-based gait) while leveraging granular individual difference annotations. By doing so, this study establishes a solid baseline for personalized depression detection that accounts for age-specific manifestations and inter-individual variability, facilitating the development of more robust and generalizable models in this evolving research direction.

A. MPDD-Young

The MPDD-Young dataset comprises data from 110 college student participants, serving as a core component of the MPDD-AVG 2026 challenge. Extending the previous MPDD dataset [13], this updated version integrates audiovisual interviews with ambulatory gait monitoring to provide a more comprehensive multimodal perspective.

Data Collection and Protocol. Participants first undergo a semi-structured interview assessing academic stress, social functioning, and emotional well-being, during which facial expressions and speech are recorded via a front-facing camera and microphone. Subsequently, participants perform a natural walking task within a designated area while equipped with wearable IMU sensors, capturing continuous gait sequences including temporal and spatial parameters (e.g., stride length, cadence) that serve as clinically relevant indicators of psychomotor symptoms.

Annotations and Supported Tasks. The dataset provides PHQ-9 severity scores, Big Five-10 personality trait evaluations, and demographic information including gender, age, and birth region. Based on these annotations, MPDD-Young supports three tasks: (1) *binary classification* distinguishing non-depressed from depressed individuals; (2) *ternary classification* categorizing participants into normal, mild, and severe depression groups; and (3) *regression* for continuous PHQ-9 score prediction. This comprehensive scheme facilitates investigation of how academic stress, social environment, and personality traits collectively contribute to depression manifestation in young adults.

B. MPDD-Elderly

The MPDD-Elderly dataset comprises 110 older adult participants, forming the geriatric cohort of the MPDD-AVG benchmark. Unlike early-onset depression, late-life depression (LLD) manifestations are heavily influenced by somatic comorbidities and socio-environmental stressors, motivating an annotation scheme that explicitly preserves granular inter-individual differences beyond monolithic behavior-to-symptom mappings.

Data Collection and Protocol. Data acquisition follows a dual-phase protocol to capture complementary behavioral phenotypes. Participants first engage in semi-structured clinical interviews, yielding episodic audio-visual affective cues. To explicitly quantify psychomotor retardation, participants subsequently perform unconstrained walking tasks equipped

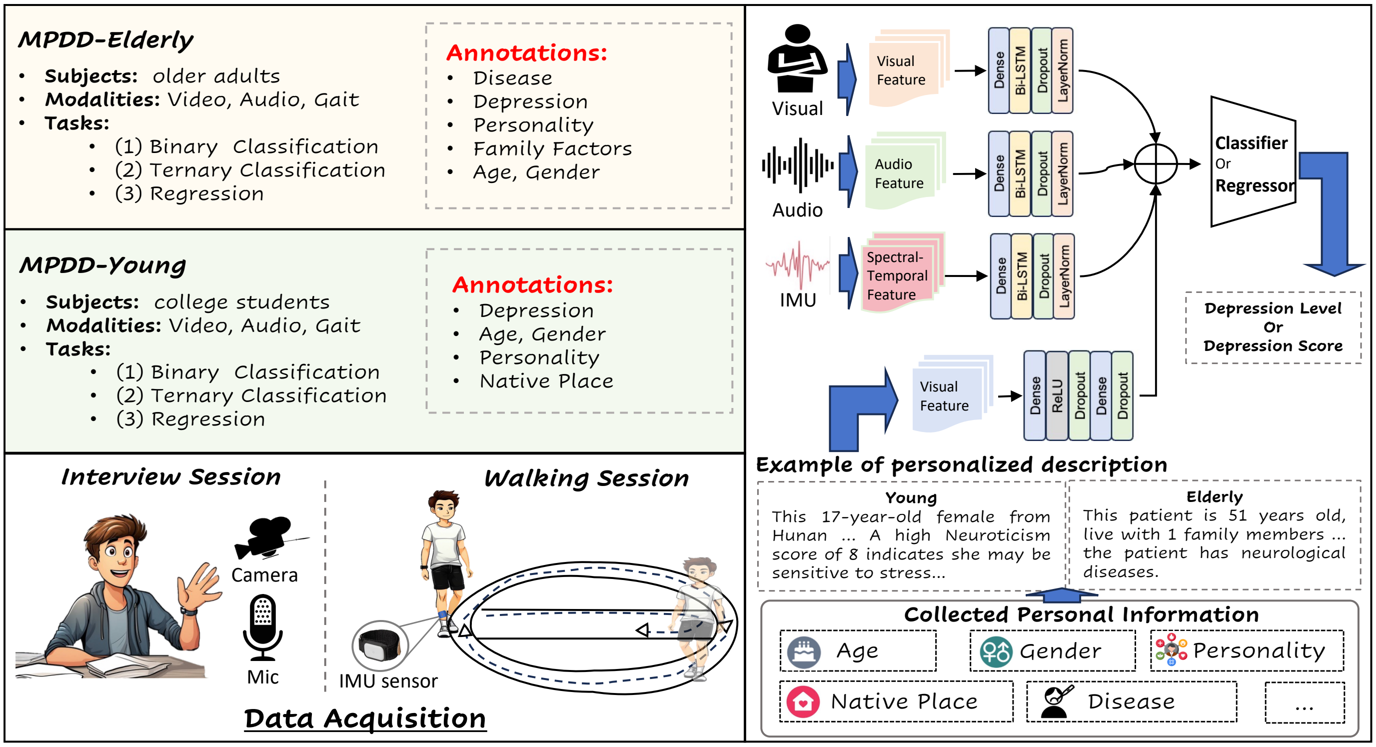

Fig. 1. Overview of the MPDD-AVG Dataset and Baseline System. (Left) The MPDD-AVG benchmark includes MPDD-Elderly (older adults) and MPDD-Young (college students), which are collected via interview sessions (video + audio) and walking sessions (IMU/gait). Each sample is annotated with depression labels, personality traits, and demographic information, supporting binary classification, ternary classification, and regression tasks. Personalized textual descriptions are generated from the collected metadata to provide subject-specific context. (Right) The baseline system extracts visual, acoustic, and spectral-temporal features from the three modalities, encodes each through a dedicated Dense–BiLSTM–Dropout–LayerNorm module, and fuses the representations to predict depression level or score.

with wearable IMU sensors. This dual-session design bridges static conversational behaviors with continuous ambulatory kinematics, enabling holistic characterization of depressive symptomatology.

Annotations and Supported Tasks. MPDD-Elderly provides multidimensional annotations to model cohort heterogeneity. Depression severity is annotated via PHQ-9 scores, supporting three tasks: (1) *binary classification* of non-depressed versus depressed individuals; (2) *ternary classification* into normal, mild, and severe groups; and (3) *regression* for continuous severity score prediction. The dataset is further enriched with Big Five-10 personality profiles, family factors including financial stress and co-resident counts, and somatic disease annotations stratifying healthy individuals against those with endocrine, circulatory, and neurological disorders. These structured annotations compel the design of architectures that explicitly condition behavioral representations on a patient's physiological and socioeconomic profile, advancing LLD assessment toward highly individualized symptom decoding. The detailed distribution of PHQ-9 score labels across the training and testing sets for both tracks is provided in Table II.

TABLE II
THE DISTRIBUTION OF DIFFERENT PHQ-9 SCORE LABELS IN TRACK 1 AND TRACK 2.

Dataset	Label	Training set	Testing set	Total
Track 1 (Elderly)	0 (Normal)	61	16	77
	1 (Mild)	16	4	20
	2 (Severe)	10	3	13
Track 2 (Young)	0 (Normal)	45	11	56
	1 (Mild)	33	8	41
	2 (Severe)	10	3	13
Total		175	45	220

III. BASELINE SYSTEM

A. Model

To establish a rigorous and reproducible benchmark for MPDD-AVG, we introduce a canonical multimodal baseline architecture, as illustrated in Fig. 1, inspired by general multimodal learning principles [15] and shared across all sub-tracks (A−V+P, A−V−G+P, and G+P). Rather than engineering heavily specialized networks for individual modality settings, this unified framework is designed to perform depression classification under a shared modeling protocol. This design effectively isolates the diagnostic contributions of different behavioral combinations and individual-difference priors, ensuring that comparisons across sub-tracks—and between the young and elderly cohorts—remain strictly controlled.

The architecture processes two fundamentally distinct streams of information. The first is modality-specific sequential evidence, encompassing episodic audio-visual dynamics extracted from semi-structured interviews and continuous ambulatory kinematics from IMU-based gait tracking. The second is participant-level contextual information, encoded as a fixed-length embedding derived from the released individual-difference attributes. For any given sub-track, the model activates only the specified input modalities while preserving a consistent processing pipeline.

To preserve modality-specific temporal dynamics before multimodal integration, each behavioral stream is encoded independently. Audio, video, and gait sequences are mapped into compact latent representations via lightweight temporal encoders. Concurrently, the participant-level embedding is transformed by a shallow projection module into a compatible contextual prior. By maintaining this consistent temporal encoding and late-fusion logic, the baseline successfully retains sequence-level dynamics in a streamlined framework, circumventing the architectural bloat typically associated with multi-stream networks. More broadly, this design is also compatible with attention-based fusion paradigms derived from Transformer architectures [16].

Upon modality-specific encoding, the resulting representations are aggregated at a high level and fed into a shallow classification head to predict depression severity. Under `A-V+P`, the model fuses conversational cues with participant context. Under `A-V-G+P`, ambulatory psychomotor evidence (Gait) is further integrated. Under `G+P`, the framework isolates the joint diagnostic yield of gait kinematics and individual differences. This unified prediction logic establishes a transparent reference for evaluating multimodal fusion strategies.

B. Feature Representations

Deviating from conventional fixed-window aggregation strategies, the released baseline adopts an **unsegmented sequence-level** feature extraction paradigm across all modalities. By retaining frame-level or sequence-level descriptors without prior windowing, the pipeline compels downstream encoders to dynamically learn global temporal dependencies. Consequently, audio and video preserve episodic conversational shifts, gait retains continuous kinematic rhythms, and individual differences are instituted as a stable subject-level context. This representation strategy aligns perfectly with the shared sequential modeling framework.

- **Acoustic Representations**: We provide three complementary feature families. MFCCs capture low-level spectral and timbral structures, offering a classical characterization of speech acoustics. OpenSMILE [17] extracts hand-crafted paralinguistic descriptors, quantifying nuanced prosodic and voice-quality patterns fundamental to affective computing. Wav2Vec2 [18] supplies deep, pretrained contextualized speech representations, complementing traditional descriptors with high-level semantic abstractions.
- **Visual Representations**: The baseline incorporates DenseNet [19], ResNet [20], and OpenFace extractors. DenseNet and ResNet provide robust frame-level appearance embeddings of the upper body and face within a generic visual space. Conversely, OpenFace explicitly targets fine-grained facial behaviors and expression-related dynamics. This configuration enables a comprehensive contrast between general-purpose spatial features and behavior-oriented facial kinematics.
- **Gait (IMU) Representations**: Serving as an objective proxy for psychomotor retardation, gait is represented by IMU-based motion sequences derived from 3D acceleration, angular velocity, and angle-related orientation signals. These sequences continuously quantify ambulatory dynamics, uniquely capturing sub-clinical movement manifestations that are structurally invisible in stationary conversational expressions.
- **Personalized Description Embeddings**: To computationally operationalize individual-difference metadata, we employ a semantic textualization strategy. Cohort-specific structured attributes (e.g., personality traits, comorbidities) are organized with a template and feed into ChatGLM to yield coherent textual descriptions, which are subsequently encoded into dense, fixed-length semantic vectors. This transforms isolated metadata fields into a stable subject-level contextual prior, explicitly compelling the model to account for inter-individual heterogeneity.

C. Personalized Features

To model individual heterogeneity, we adopt the personalized feature extraction pipeline established in the previous MPDD Challenge [13]. This approach leverages participant-specific attributes, including Big Five-10 personality traits, financial stress levels, and gait-derived psychomotor signatures, to generate individualized representations.

Specifically, we transform these structured attributes into natural language descriptions via a generative language model (ChatGLM3) [21]. These descriptions are then encoded using the RoBERTa-large model [22], from which we extract the hidden state of the first token to obtain a 1024-dimensional embedding. These embeddings serve as compact, semantically-rich personalized features, which are subsequently concatenated with acoustic, visual, and gait features for downstream classification.

IV. Metrics

To comprehensively evaluate the performance of our proposed baseline model, we employ a set of standard metrics for both classification and regression tasks. Specifically, we utilize Accuracy, Macro-F1, and Cohen's Kappa for classification, and Concordance Correlation Coefficient (CCC), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) for regression. Given the potential class imbalance in the dataset, Macro-F1 is reported together with Accuracy and Cohen's Kappa to ensure a robust assessment of the model's capabilities across different depression severity levels.

Accuracy measures the proportion of correct predictions:

$$\text{Acc} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Samples}} \tag{1}$$

The F1-score, as the harmonic mean of precision and recall, is particularly vital for addressing class imbalance frequently encountered in mental health datasets:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

where

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

Macro-F1 computes the F1-score for each class independently and takes the arithmetic mean, ensuring that all depression severity levels contribute equally to the final metric irrespective of their sample frequency:

$$\text{Macro-F}_1 = \frac{1}{C} \sum_{i=1}^{C} F_1^{(i)} \tag{4}$$

where C is the number of classes.

Cohen's Kappa quantifies inter-rater agreement by correcting for chance, providing a robust measure for evaluating clinical diagnostic consistency:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{5}$$

where p_o is the observed agreement and p_e is the expected agreement by chance.

For the depression severity prediction task, we employ the following metrics calculated based on the log-transformed scores y and $\hat{y}$. Specifically, the Root Mean Squared Error (RMSE) quantifies the magnitude of prediction errors:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2} \tag{6}$$

Mean Absolute Error (MAE) provides an interpretable average prediction error:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{7}$$

The Concordance Correlation Coefficient (CCC) evaluates the agreement between the transformed predictions and labels by combining measures of precision and accuracy [23]:

$$\text{CCC} = \frac{2\rho\, \sigma_y\, \sigma_{\hat{y}}}{\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2} \tag{8}$$

where ρ is Pearson's correlation coefficient, σ denotes the standard deviation, and μ denotes the mean.

Note that, for the PHQ-9 score prediction task, to mitigate the effect of skewed data distribution and stabilize the training process, we perform a $\log(1+x)$ transformation on all original scores. Throughout the remainder of this paper, y and $\hat{y}$ denote the log-transformed ground-truth and predicted values, respectively:

$$y = \ln(1 + x_{raw}), \quad \hat{y} = \ln(1 + \hat{x}_{raw}) \tag{9}$$

where x_{raw} represents the original scale score (0–27). Consequently, all regression evaluation metrics, including CCC, RMSE, and MAE, are computed within this logarithmic space. Together, these metrics provide a comprehensive evaluation of classification performance, regression accuracy, and prediction consistency for the proposed baseline across different challenge settings.

V. Results

The baseline results for the Elderly and Young tracks are summarized in Table III. Overall, the results show that incorporating gait information is beneficial for depression recognition, especially in binary classification, while the relative advantages of different modality combinations vary across age groups.

For the Elderly track, adding gait information to the audio-visual-personality setting improves binary classification performance, with F1 increasing from 0.6247 to 0.7319 and ACC from 0.6957 to 0.7826. A similar tendency can also be observed in the Young track, where the binary F1 rises from 0.4762 to 0.5832 after introducing gait. These results suggest that gait provides complementary behavioral cues beyond conventional audio-visual signals and is helpful for depression detection in both cohorts.

At the same time, the best-performing configuration differs between the two age groups. In the Young track, the gait-only setting with personality (G+P) achieves the strongest binary performance, reaching an F1 of 0.6364 and an ACC of 0.6364, which indicates that gait-related behavioral patterns alone can already provide strong discriminative information for younger participants. In contrast, for the Elderly track, G+P obtains a lower F1 of 0.4103 than A-V+P and A-V-G+P, although its ACC remains comparable to A-V+P. This difference suggests that elderly subjects may benefit more from multimodal fusion, whereas gait appears to be particularly informative for the Young group.

Across all sub-tracks, ternary classification remains noticeably more difficult than binary classification. For example, in the Elderly A-V-G+P setting, the F1 score decreases from 0.7319 in the binary task to 0.4140 in the ternary task. Similar gaps can be observed in other settings, indicating that fine-grained severity discrimination is substantially more challenging than binary depression recognition.

It is also worth noting that regression metrics do not always follow the same trend as classification metrics. Since CCC, RMSE, and MAE are evaluated in the log-transformed space ($\ln(1+x)$), their variations may reflect different aspects of prediction behavior from those captured by F1 and ACC. Overall, the results highlight the value of gait as an effective behavioral modality and reveal cohort-dependent differences in the relative advantages of different modality combinations.

VI. Conclusion

In this work, we present the official baseline system for the MPDD-AVG 2026 Challenge, introducing a comprehensive benchmark for personality-aware, multimodal depression detection across two age cohorts. By jointly integrating audio-visual interview cues, continuous gait kinematics captured via wearable IMU sensors, and granular individual-difference

TABLE III
BASELINE RESULTS ON THE MPDD-AVG DATASET (TRACK 1: ELDERLY, TRACK 2: YOUNG). REPRESENTATIVE BASELINE RESULTS ARE REPORTED.

Track	**Sub-track**	**A/V Features**	**Task**	F1	ACC	Kappa	CCC	RMSE	MAE
Elderly	A-V+P	opensmile + resnet	Binary	0.6247	0.6957	0.2512	0.1463	0.9650	0.8138
Elderly	A-V+P	mfcc + resnet	Ternary	0.4162	0.6957	0.3090	0.1093	1.1027	0.9273
Elderly	A-V-G+P	mfcc + resnet	Binary	0.7319	0.7826	0.4651	0.1527	1.0499	0.8861
Elderly	A-V-G+P	opensmile + densenet	Ternary	0.4140	0.7391	0.2023	0.0948	0.8913	0.7427
Elderly	G+P	-	Binary	0.4103	0.6957	0.0000	0.0207	0.8812	0.7159
Elderly	G+P	-	Ternary	0.2735	0.6957	0.0000	0.0111	0.8784	0.7236
Young	A-V+P	wav2vec + openface	Binary	0.4762	0.5455	0.0909	0.1428	0.8461	0.6855
Young	A-V+P	wav2vec + openface	Ternary	0.3235	0.4545	0.0186	0.0747	0.8144	0.6028
Young	A-V-G+P	wav2vec + openface	Binary	0.5832	0.5909	0.1818	0.1718	0.7876	0.6267
Young	A-V-G+P	wav2vec + resnet	Ternary	0.3254	0.4545	0.0504	0.1746	0.7464	0.6154
Young	G+P	-	Binary	0.6364	0.6364	0.2727	0.3291	0.7002	0.5321
Young	G+P	-	Ternary	0.3651	0.5000	0.1479	0.2321	0.7739	0.6058

Note: Regression metrics (CCC, RMSE, and MAE) are reported in the log-transformed space ($\ln(1 + x)$).

annotations including personality traits and demographic attributes, the MPDD-AVG dataset directly addresses two critical limitations of existing benchmarks: the neglect of psychomotor behavioral signatures and the absence of explicit individual heterogeneity modeling. Our baseline establishes transparent and reproducible reference performance across all three sub-tracks (A-V+P, A-V-G+P, and G+P) under both PHQ-9 regression and binary/ternary classification settings. Empirical results consistently demonstrate that gait incorporation yields clear diagnostic gains over audio-visual-only baselines in several settings, while the proposed framework further enables personality-conditioned and individualized depression modeling across both cohorts. These findings underscore the clinical and computational value of extending depression detection beyond static conversational observations toward a holistic, personalized multimodal paradigm. We hope MPDD-AVG serves as a foundational testbed to advance automated depression assessment research, encouraging the community to develop models that are more sensitive to psychomotor symptoms and better able to account for inter-individual variability in depressive symptomatology.

ACKNOWLEDGMENT

This research is supported by the National Natural Science Foundation of China (Grant No. 62306068) Project, the Natural Science Foundation of Hebei Province, China (Grant No. F2024501002), and the Fundamental Research Funds for the Central Universities (Grant No. N2523005).

REFERENCES

[1] World Health Organization, “Depression and other common mental disorders: Global health estimates,” 2017, wHO/MSD/MER/2017.2. [Online]. Available: https://iris.who.int/handle/10665/254610

[2] F. Ringeval, B. Schuller, M. Valstar, N. Cummins, R. Cowie, L. Tavabi, M. Schmitt, S. Alisamir, S. Amiriparian, E. Messner, S. Song, S. Liu, Z. Zhao, A. Mallol-Ragolta, Z. Ren, M. Soleymani, and M. Pantic, “AVEC 2019 workshop and challenge: State-of-mind, detecting depression with AI, and cross-cultural affect recognition,” in *Proceedings of the 9th International Audio/Visual Emotion Challenge and Workshop*, 2019, pp. 3–12.

[3] J. Gratch, R. Artstein, G. Lucas, G. Stratou, S. Scherer, A. Nazarian, R. Wood, J. Boberg, D. DeVault, S. Marsella, D. Traum, A. Rizzo, and L. Morency, “The distress analysis interview corpus of human and computer interviews,” in *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC’14)*. Reykjavik, Iceland: European Language Resources Association (ELRA), 2014.

[4] D. Bennabi, P. Vandel, C. Papaxanthis, T. Pozzo, and E. Haffen, “Psychomotor retardation in depression: A systematic review of diagnostic, pathophysiologic, and therapeutic implications,” *BioMed Research International*, vol. 2013, p. 158746, 2013.

[5] S. Radovanović, M. Jovičić, N. P. Marić, and V. Kostić, “Gait characteristics in patients with major depression performing cognitive and motor tasks while walking,” *Psychiatry Research*, vol. 217, no. 1–2, pp. 39–46, 2014.

[6] D. N. Klein, R. Kotov, and S. J. Bufferd, “Personality and depression: Explanatory models and review of the evidence,” *Annual Review of Clinical Psychology*, vol. 7, pp. 269–295, 2011.

[7] S. Moussavi, S. Chatterji, E. Verdes, A. Tandon, V. Patel, and B. Ustun, “Depression, chronic diseases, and decrements in health: Results from the world health surveys,” *The Lancet*, vol. 370, no. 9590, pp. 851–858, 2007.

[8] A. Fiske, J. L. Wetherell, and M. Gatz, “Depression in older adults,” *Annual Review of Clinical Psychology*, vol. 5, pp. 363–389, 2009.

[9] H. Dibeklioğlu, Z. Hammal, and J. F. Cohn, “Dynamic multimodal measurement of depression severity using deep autoencoding,” *IEEE Journal of Biomedical and Health Informatics*, vol. 22, no. 2, pp. 525–536, 2018.

[10] J. Yoon, C. Kang, S. Kim, and J. Han, “D-Vlog: Multimodal vlog dataset for depression detection,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 11, 2022, pp. 12 226–12 234.

[11] H. Cai, Z. Yuan, Y. Gao, S. Sun, N. Li, F. Tian, H. Xiao, J. Li, Z. Yang, X. Li, Q. Zhao, Z. Liu, Z. Yao, M. Yang, H. Peng, J. Zhu, X. Zhang, G. Gao, F. Zheng, R. Li, Z. Guo, R. Ma, J. Yang, L. Zhang, X. Hu, Y. Li, and B. Hu, “A multi-modal open dataset for mental-disorder analysis,” *Scientific Data*, vol. 9, no. 1, p. 178, 2022.

[12] J. A. Miranda-Correa, M. K. Abadi, N. Sebe, and I. Patras, “AMIGOS: A dataset for affect, personality and mood research on individuals and groups,” *IEEE Transactions on Affective Computing*, vol. 12, no. 2, pp. 479–493, 2021.

[13] C. Fu, Z. Fu, Q. Zhang, X. Kuang, J. Dong, K. Su, Y. Su, W. Shi, J. Yao, Y. Zhao, S. Zhao, J. Wang, S. Song, C. Liu, Y. Yoshikawa, B. Schuller, and H. Ishiguro, “The first MPDD challenge: Multimodal personality-aware depression detection,” in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 13 924–13 929.

[14] B. Zou, J. Han, Y. Wang, R. Liu, S. Zhao, L. Feng, X. Lyu, and H. Ma, “Semi-structural interview-based chinese multimodal depression corpus towards automatic preliminary screening of depressive disorders,” *IEEE Transactions on Affective Computing*, vol. 14, no. 4, pp. 2823–2838, 2023.

[15] T. Baltrušaitis, C. Ahuja, and L. Morency, “Multimodal machine learn-

ing: A survey and taxonomy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019.

[16] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017, pp. 5998–6008.

[17] F. Eyben, M. Wöllmer, and B. Schuller, "openSMILE: The munich versatile and fast open-source audio feature extractor," in *Proceedings of the 18th ACM International Conference on Multimedia*, 2010, pp. 1459–1462.

[18] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020.

[19] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.

[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[21] A. Zeng, X. Liu, Z. Du, Z. Wang, H. Lai, M. Ding, Z. Yang, Y. Xu, W. Zheng, X. Xia, W. L. Tam, Z. Ma, Y. Xue, J. Zhai, W. Chen, Z. Liu, P. Zhang, Y. Dong, and J. Tang, "GLM-130b: An open bilingual pre-trained model," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=-Aw0rrrPUF

[22] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," *arXiv preprint arXiv:1907.11692*, 2019. [Online]. Available: https://arxiv.org/abs/1907.11692

[23] L. I.-K. Lin, "A concordance correlation coefficient to evaluate reproducibility," *Biometrics*, vol. 45, no. 1, pp. 255–268, 1989.